

Logging

Version: 1.0

Purpose

This document defines the logging architecture of Trading Platform Pro.

Logging is a core infrastructure service used to improve observability, diagnostics, troubleshooting and operational transparency.

Objectives

The logging system shall provide:

- structured logging
- deterministic output
- consistent formatting
- configurable log levels
- centralized configuration
- minimal runtime overhead

Design Principles

Logging should be:

- consistent
- predictable
- informative
- lightweight

Log messages should explain **what happened** and provide sufficient context for diagnosis.

Log Levels

Supported log levels:

Level	Purpose
DEBUG	Detailed diagnostic information
INFO	Normal operational events
WARNING	Unexpected but recoverable situations
ERROR	Operation failed
CRITICAL	System stability is affected

Logging Rules

Log when:

- services start or stop
 - configuration is loaded
 - external resources are initialized
 - significant state changes occur
 - recoverable errors happen
 - unexpected exceptions occur
- Do **not** log trivial implementation details.

Message Style

Log messages should be:

- short
- descriptive
- consistent

Example:

Configuration loaded successfully.

Avoid:

Entered function xyz...

Structured Logging

Whenever practical, include structured context:

- component
- operation
- identifier
- execution time
- correlation id

Avoid embedding complex data inside free-form text.

Exception Logging

Exceptions should:

- be logged once
- include the root cause
- preserve the stack trace

Avoid duplicate logging of the same exception.

Sensitive Data

Never log:

- passwords
- API keys
- authentication tokens
- broker credentials
- personal information

Sensitive values must always be masked or omitted.

Performance

Logging should:

- avoid unnecessary string formatting
- avoid excessive allocations
- remain efficient in high-frequency code paths

Configuration

Logging configuration is managed centrally.

Typical configuration includes:

- log level
- output destination
- formatting
- file rotation

- retention policy

Testing

Logging behaviour should be testable where appropriate.
Critical logging paths should be covered by automated tests.

Log Retention

Log retention shall be configurable.

Retention policies should define:

- retention period
- maximum file size
- rotation strategy
- archival rules

Log Review Checklist

Before release verify:

- log levels are appropriate
- no duplicate logging
- no sensitive information logged
- structured context included where useful
- logging remains performant

Audit Logging

Security and business-critical events should produce audit logs.

Examples:

- authentication events
- configuration changes
- order lifecycle events
- permission changes

Audit logs should be immutable where practical.

Related Documents

- Runtime.md
- Configuration.md
- Monitoring.md
- Architecture.md
- AGENTS.md